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SYSTEM AND METHOD FOR TEMPERATURE DETERMINATION

Abstract

In variants, an automatically-identifiable temperature probe can include: a probe body, one or more sensors, a connector, and/or any other suitable components. In variants, the method for temperature determination can include: determining a set of electrical signals, determining a temperature probe type based on the set of electrical signals, determining a sensor resolution model based on the temperature probe type, and determining a set of final temperature estimates based on the set of electrical signals and the sensor resolution model.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/135,967, filed on Apr. 18, 2023, which is a continuation of U.S. application Ser. No. 17/889,700, filed on Aug. 17, 2022, both of which are incorporated in their entireties by reference.

TECHNICAL FIELD

[0002] This invention relates generally to the cooking field, and more specifically to a new and useful temperature determination system and method in the cooking field.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0003] FIG. 1 is a schematic representation of the method.

[0004] FIG. 2 is a schematic representation of the system.

[0005] FIG. 3 is an illustrative example of a variant of the method.

[0006] FIG. 4 is an illustrative example of using the temperature probe with a cooking system.

[0007] FIG. 5 is an illustrative example of wirelessly connecting the cooking system to a remote computing system and/or a user device.

[0008] FIGS. 6A-6C are illustrative examples of connectors of the temperature probe.

[0009] FIG. 7 is an illustrative example of mappings between ground-signal conductor patterns and temperature probe types.

[0010] FIG. 8 is an illustrative example of a set of models associated with different temperature probe types.

[0011] FIG. 9 is an illustrative example of a variant of the method.

[0012] FIG. 10 is an illustrative example of using the temperature probe with a cooking system.

[0013] FIG. 11 is an illustrative example of using the temperature probe with a cooking system.

[0014] FIG. 12 is an illustrative example of a set of models associated with different temperature probe types.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0015] The following description of the preferred embodiments of the invention is not intended to limit the invention to these preferred embodiments, but rather to enable any person skilled in the art to make and use this invention.

1. OVERVIEW

[0016] In variants, the temperature probe **100** can include: a probe body **110**, one or more sensors **120**, a probe connector **130**, and/or any other suitable components. In examples, the connector can include a live/grounded conductor pattern that is specific to the temperature probe type, such that a cooking system **200** can use the connector readouts to both identify the temperature probe type and determine a temperature measurement of a measured substance.

[0017] As shown in FIG. 2, the temperature probe can be used with: one or more cooking systems **200**, one or more remote computing systems **300**, and/or any other suitable components.

[0018] As shown in FIG. 1, the method can include: determining a set of electrical signals **S100**,

determining a temperature probe type **S200**, determining a final temperature estimate **S300**, and/or any other suitable elements.

[0019] The system and/or method functions to identify the temperature probe **100** connected to a cooking system **200**, retrieve the temperature resolution model for the identified temperature probe **100**, and/or determine one or more temperature estimates for an object (e.g., based on the temperature resolution model). The object is preferably food (e.g., meat, vegetable, soup, candy, baked goods, etc.), but can additionally and/or alternatively be a human body, an animal body, matter (e.g., air, water, soil, substance, etc.), and/or any other suitable object. The temperature estimate can be: internal object temperature, external object temperature, ambient temperature, and/or any other suitable temperature.

2. BENEFITS

[0020] The cooking system can confer several benefits over conventional systems.

[0021] Variants of the technology can automatically identify a temperature probe type at a cooking system (e.g., oven, grill, smoker, temperature probe hub, etc.) without user selection. In these variants, the cooking system can receive a set of electrical signals from the temperature probe and determine a temperature probe type, based on which conductor(s) of the temperature probe are associated with the set of sensed electrical signals. In an illustrative example, the technology can use the pattern of grounded and signal conductors to uniquely identify a temperature probe type and/or make and model. In examples, this can be particularly beneficial since the same resistance measured by different temperature probe types and/or manufacture can map to different temperatures. By automatically identifying the temperature probe type and/or manufacture, the technology can enable the cooking system to generate more accurate temperature estimates for the food by selecting the correct temperature model for temperature probe signal interpretation. This can also create a more facile user experience, since the user no longer needs to manually enter the probe type into the cooking system each time the probe is turned on or plugged in.

[0022] However, the systems and/or method can confer any suitable set of benefits.

3. SYSTEM

[0023] The temperature probe **100** functions to measure one or more temperatures of an object. The system can include one temperature probe **100**, multiple temperature probes **100**, and/or any other suitable number of temperature probes. The temperature probe **100** preferably measures an internal temperature of an object, but can additionally and/or alternatively measure an external temperature of an object. The temperature probe can be configured to be: continuously operable (e.g., from a continuous insertion of the temperature probe **100** into an object), intermittently operable (e.g., from an intermittent insertion of the temperature probe **100** into an object), and/or otherwise configured for use. The temperature probe **100** is preferably removably coupleable to a cooking system **200**, but can additionally and/or alternatively be mounted to a cooking system **200**, and/or otherwise coupled to a cooking system. The temperature probe **100** is preferably physically plugged into the cooking system **200** (e.g., connected to the cooking system **200** by a wired connection), but can additionally or alternatively be connected by a wireless connection and/or any other suitable connection. The temperature probe **100** can be: automatically actuated, manually actuated, passively actuated, and/or otherwise actuated. The temperature probe **100** can be grounded or ungrounded.

[0024] The temperature probe **100** can include: a probe body **110**, one or more sensors **120**, a probe connector **130**, optionally a power source **140**, and/or any other suitable components.

[0025] The probe body **110** functions to support the one or more sensors **120**. The temperature probe **100** can include one probe body **110**, multiple probe bodies **110**, and/or any other suitable number of probe bodies. The probe body **110** is preferably rigid, but can alternatively be flexible. The probe body **110** is preferably straight, but can alternatively be curved and/or have any other suitable geometry. The probe body **110** is preferably thermally conductive, but can alternatively be thermally insulative, thermally dissipative, and/or any other suitable thermal properties. The probe

body **110** is preferably metal (e.g., aluminum, stainless steel, copper, etc.), but can additionally and/or alternatively be ceramic, polymer (e.g., silicone rubber, plastic, etc.), glass, and/or any other suitable material. The first end of the probe body **110** is preferably statically connected to the probe connector **130** (e.g., via a wire, solder, and/or other physical connection), but can additionally or alternatively be wirelessly connected to the probe connector **130**, and/or otherwise coupled to the connector and/or cooking system **200**. The probe body **110** and/or second end thereof (e.g., probe tip, opposing end, free end, etc.) preferably supports (e.g., mounts) one or more sensors **120**. However, the probe body **110** can be otherwise configured.

[0026] The sensors **120** function to determine temperature measurements. The temperature probe **100** can include: one sensor **120**, multiple sensors **120**, and/or any other suitable number of sensors. The sensors **120** on each temperature probe **100** can be of the same or different sensor type or manufacture. The sensors are preferably analog sensors (e.g., output a resistance, voltage, or current readout), but can additionally or alternatively be a digital sensor (e.g., output a digital stream of measurements). The sensors are preferably temperature sensors, but can alternatively be humidity sensors, particle sensors, light sensors, motion sensors (e.g., IMU, accelerometers, etc.), pressure sensors, and/or a sensor for any other sensing modality. The temperature sensors can be: thermistors, temperature detectors (e.g., resistance temperature detectors), thermocouples, semiconductor-based sensors and/or any other suitable sensor. Examples of temperature sensors that can be used include: resistance temperature detectors (RTDs), negative temperature coefficient thermistors (NTCs), positive temperature coefficient thermistors (PTCs), and/or any other suitable sensors. The temperature probe **100** and/or each sensor **120** thereof can have a measurement accuracy and/or measurement precision within: 0.1° F., 0.2° F., 0.5° F., 1° F., 1.5° F., 2° F., 10° F., more than 10° F., within any range bounded by any of the aforementioned values, and/or any other suitable measurement accuracy and/or measurement precision. The temperature probe **100** and/or each sensor **120** can have a measurement range bounded by any of the following values: -1000° F., -500° F., -200° F., -100° F., -50° F., 0° F., 20° F., 100° F., 300° F., 600° F., 4000° F., and/or any other suitable measurement range. The sensors **120** can have: an exponential response, a linear resistance response, a voltage response, a hysteretic response, and/or any other suitable response. The sensors **120** preferably output varying resistance indicative of different measured temperatures, but can additionally or alternatively output: voltage, current, and/or vary any other suitable electromagnetic parameter value as a function of the measured temperature.

[0027] Each temperature probe **100** can include one or more sensors **120**. In variants, the temperature probe **100** can include multiple sensors **120** to confer resilience to common user errors. Such variants can provide flexibility in user error resulting from improper temperature probe insertion, such as common errors associated with: the probe tip not touching the center of the object (e.g., food), the temperature probe **100** touching the walls of a container (e.g., of a cooking system **200**), and/or any other additional challenges/common errors. The multiple sensors **120** can be of the same or different type of sensor. The multiple measurements sampled by the multiple sensors can be: fused into a single measurement (e.g., by taking the mean, median, minimum, maximum, most stable, least deviation, and/or any other measurement, etc.), used to determine a spatial temperature distribution, used to determine a timeseries of temperatures (e.g., historic and/or future temperature measurement values), and/or otherwise used. In an example, determining the lowest measured temperature between the multiple sensors **120** can achieve a more accurate temperature estimation of internal temperature of the object.

[0028] Each sensor **120** can be: flush with the probe body **110**, recessed from the probe body **110**, protrude from the probe body **110**, aligned along the probe body **110**, inside the probe body **110**, and/or otherwise arranged relative to the probe body. In an example, multiple sensors **120** are linearly aligned within the probe body **110** (e.g., along the same side, along the longitudinal axis, etc.). In another example, multiple sensors **120** are located at different positions along the probe body length and arranged in a pattern (e.g., a spiral pattern, a zigzag pattern, etc.). However,

multiple sensors **120** can be otherwise arranged. The separation distance between multiple sensors **120** can be: 1 mm, 5 mm, 10 mm, 15 mm, 20 mm, within any range bounded by any of the aforementioned values, and/or any other suitable distance.

[0029] Measurements (e.g., temperature measurements) are preferably contemporaneously (e.g., concurrently, within the same timeframe, etc.) sampled by the multiple sensors **120**, but can additionally and/or alternatively be asynchronously sampled. In variants, a temperature estimate can be determined (e.g., by a processing system **250** of a cooking system **200**, a remote computing system **300**) based on the temperature measurements sampled by multiple sensors **120**, but can additionally and/or alternatively be based on one sampled temperature measurement by a single sensor **120** (e.g., a timeseries of temperature measurements from the sensor **120**), and/or otherwise determined.

[0030] However, the sensors **120** can be otherwise configured.

[0031] The probe connector **130** functions to connect a cooking system **200** to the temperature probe **100**; examples shown in FIG. 4, FIG. 10, and FIG. 11. The probe connector **130** preferably removably connects a cooking system **200** to the temperature probe **100**, but can alternatively permanently connect the cooking system **200** to the temperature probe **100**, and/or otherwise configured. The probe connector **130** can be: a mechanical connector, a data connector, a power connector, and/or any other suitable connector. The temperature probe **100** can include: one probe connector **130**, multiple probe connectors **130** (e.g., 2, 3, etc.), and/or any other suitable number of connectors. The probe connector **130** is preferably electrically conductive, but can alternatively be electrically insulative, thermally dissipative, and/or have any other suitable properties. The probe connector **130** can be: metal (e.g., aluminum, stainless steel, copper, etc.), ceramic, polymer (e.g., silicone rubber, plastic, etc.), glass, and/or any other suitable material. The probe connector **130** is preferably a male connector, but can additionally and/or alternatively be a female connector. The probe connector **130** preferably connects to a female connector, but can additionally and/or alternatively connect to a male connector. The probe connector **130** can be: a plug or socket connector, a crimp on connector, a soldered connector, a binding post, a screw terminal, a ring and spade connector, a blade connector, twist-on wire connector, alligator clip, and/or any other suitable connector. The probe connector **130** is preferably an auxiliary plug, but can additionally and/or alternatively be a USB-A cable, a USB-C cable, a micro-USB cable, a headphone plug, and/or any other suitable connector.

[0032] The probe connector **130** preferably includes two or more conductors, but can additionally and/or alternatively include one conductor, and/or any other suitable number of conductors. Each conductor of the probe connector **130** is preferably electrically connected (e.g., wired) to a sensor **120** of the temperature probe **100**, but can additionally and/or alternatively be electrically connected to multiple sensors **120** of the temperature probe **100** or no sensors of the temperature probe **100** (e.g., be grounded to a housing or other electrical ground). The signal carried (e.g., transmitted, received, etc.) by each conductor is preferably directly output by the respective sensor, but can additionally or alternatively be otherwise processed. The signal is preferably an analog signal, but can alternatively be a digital signal. Each conductor can be: a tip, a ring, a sleeve, and/or any other suitable component. In a first example, the probe connector **130** includes only a tip, a ring, and a sleeve (e.g., be a TRS connector; example shown in FIG. 6B). In a second example, the probe connector **130** only includes a tip, two rings, and a sleeve (e.g., be a TRRS connector; example shown in FIG. 6C). In a third example, the probe connector **130** only includes a tip, three rings, and a sleeve (e.g., be a TRRRS connector). In a fourth example, the probe connector **130** only includes a tip and a sleeve (e.g., be a TS connector; example shown in FIG. 6A). However, the probe connector **130** can include any other number of rings, tips, and/or sleeves.

[0033] In a first variant, the probe connector **130** can be permanently wired to the one or more sensors **120**.

[0034] In a second variant, the probe connector **130** can be removably coupled to the one or more

sensors **120**. For example, an adapter is used to convert a third-party temperature probe ground-signal conductor pattern to the system-standard ground-signal conductor pattern. The adapter is preferably specific to the temperature probe type and the third-party temperature probe, but can alternatively be generic. The adapter includes a female adapter connector and a male adapter connector, wherein the male adapter connector is removably coupleable to cooking system connector **240**, and the female connector is removably coupleable to the third-party connector. In an example, the female adapter connector can include a set of contacts configured to connect to the conductors of the third-party temperature probe. The set of contacts can include a contact for each third-party temperature probe conductor (e.g., one contact for each tip, ring, and/or sleeve), a single contact for multiple third-party temperature probe conductors (e.g., wherein sequential third-party temperature probe's conductors have the same signal or ground), and/or any other suitable cardinality between the third party temperature probe conductors and the contacts. The adapter can include wiring that connects the conductors of the third-party connector for the third-party probe to the conductors of the cooking system connector **240** for the system-standard temperature probe type, or be otherwise configured. For example, when a third party NTC has a grounded tip and sleeve, with a live ring, the adapter can include a female adapter connector with a tip, ring, and sleeve contacts (e.g., corresponding to the tip, ring, and sleeve positions), the male adapter connector can include a tip conductor connected to the ring contact, one or more rings connected to the tip and/or sleeve contact, and a sleeve connected to electrical ground and/or the tip and/or sleeve contact. However, the adapter can be otherwise configured.

[0035] However, the probe connector **130** can be otherwise configured.

[0036] The temperature probe **100** is preferably associated with a temperature probe type, but can additionally and/or alternatively be associated with, a temperature probe make and/or model (e.g., PT100 RTD), and/or any other suitable identifier. The temperature probe type can be associated with a number of sensors (e.g., of the sensor(s) **120**), a type of sensors (e.g., of the sensor(s) **120**), and/or any other suitable probe attribute. In examples, the temperature probe type can be: 1-sensor NTC, 2-sensor NTC, 3-sensor NTC, 1-sensor RTD, 2-sensor RTD, 3-sensor RTD, and/or any other suitable temperature probe type.

[0037] Each temperature probe, temperature probe type, or temperature sensor can be associated with a different ground-signal conductor pattern (e.g., different spatial patterns, electrical patterns, live/ground pattern, etc.), but can alternatively have the same ground-signal conductor pattern, permutation, and/or combination. A ground-signal conductor pattern is preferably unique to a temperature probe type or temperature sensor type, but can alternatively be nonunique, and/or otherwise related to the temperature probe. In a first embodiment, the ground-signal conductor pattern can be used to identify the sensor probe type (e.g., temperature probe type), which, in turn, can be used to retrieve the sensor resolution model (e.g., temperature resolution model) used to resolve sensor probe signals into sensor measurements (e.g., temperature measurements) and/or retrieve the fusion model used to fuse measurements from multiple sensors into a single readout. In a second embodiment, the ground-signal conductor pattern can be used to identify the sensor type (e.g., temperature sensor type, such as NTC or RTD), which can be used to retrieve the sensor resolution model for the sensor type (e.g., wherein the sensor resolution model can be applied to a subset or all of the signals output by the temperature probe). For example, NTC sensors can use an exponential resolution model (e.g., relating a resistance response to a temperature value using an exponential model), RTD sensors can use a linear resolution model (e.g., relating a resistance response to a temperature value using a linear model), and thermocouples can use a thermocouple-specific resolution model (e.g., relating a voltage response to a temperature value). In this embodiment, variants of the system only need to store sensor resolution models for each sensor type (e.g., does not include models for each temperature probe type or sensor combination); alternatively, the system can include sensor resolution models for: each temperature probe type, each temperature probe type and each sensor type, and/or any other suitable set of sensor resolution

models. However, the ground-signal pattern can be otherwise used. The ground-signal conductor pattern is preferably defined by whether each conductor (e.g., a tip, a ring, a sleeve, etc.) of the probe connector **130** is associated with (e.g., conducts) an electrical signal or is grounded, but can additionally and/or alternatively include any other suitable information.

[0038] In a first example, a ground-signal-ground TRS pattern can identify a 1-sensor RTD. In a second example, a ground-signal-signal-ground TRRS pattern can identify a 2-sensor RTD. In a third example, a grounded tip can identify an RTD temperature sensor. In a fourth example, a signal-ground TS pattern can identify a 1-sensor NTC. In a fifth example, a signal-signal-ground TS pattern can identify a 2-sensor NTC. In a sixth example, a signal-signal-signal-ground TS pattern can identify a 3-sensor NTC. In a seventh example, a signal detected at the tip can identify an NTC sensor. Alternatively, a grounded tip can identify a thermistor, NTC, or other temperature sensor or probe; and/or a signaled tip can identify a RTD, thermistor, or other temperature sensor or probe. However, any other suitable ground-signal pattern can identify any other suitable temperature probe type or sensor type.

[0039] The temperature probe **100** and/or sensors thereof can optionally include a power source **140** (e.g., be powered or unpowered; active or passive). The temperature probe **100** can be powered by: the power source **140** (e.g., located onboard the temperature probe **100**), a cooking system **200** (e.g., via probe connector **130** and cooking system connector **240**), and/or otherwise powered. The power source **140** is preferably a rechargeable battery, but can alternatively be a non-rechargeable battery (e.g., a disposal battery), induction coil, RF charger, and/or any other suitable power source. The power source **140** can be: a lithium-ion battery, a nickel-cadmium battery, a nickel-metal-hydride battery, a lithium-ion polymer battery, and/or any other suitable rechargeable battery. The power source **140** is preferably charged via a connector (e.g., same or different as probe connector **130**), but can alternatively be charged via Bluetooth, induction, and/or any other suitable power charging component.

[0040] However, the power source **140** can be otherwise configured.

[0041] However, the temperature probe **100** can be otherwise configured.

[0042] The temperature probe can optionally be used with: one or more cooking systems **200**, optionally one or more remote computing systems **300**, and/or any other suitable components.

[0043] The cooking system **200** functions to removably couple the temperature probe **100**, enable temperature determination using the temperature probe **100**, and/or have any other suitable functionality.

[0044] The cooking system **200** is preferably a cooking appliance (e.g., an oven, a grill, a smoker, a sous-vide pot, a refrigerator, examples shown in FIG. **4** and FIG. **10**, etc.), but can additionally and/or alternatively be a temperature probe hub (e.g., with a user interface and a set of ports or cooking system connectors **240**; example shown in FIG. **11** etc.), and/or any other suitable system. The cooking appliance can include: a cooking cavity (e.g., a heated cavity), a cooking surface (e.g., a heated cook surface such as a cooktop), cooking elements operably connected to and/or mounted within the cooking cavity and/or on the cooking surface (e.g., heating elements, convection elements, steaming elements, smoking elements, etc.), optionally sensing elements operably connected to and/or mounted within the cooking cavity and/or on the cooking surface (e.g., cameras, temperature sensors, humidity sensors, weight sensors, etc.), and/or any other suitable component.

[0045] The cooking system **200** is preferably a digitally controllable appliance, but can additionally and/or alternatively be a manually controlled appliance. The cooking system **200** can be concurrently, contemporaneously, and/or asynchronously communicatively connected to one or more: wireless networks, different cooking systems **200**, temperature probes **100**, remote computing systems **300**, and/or any other suitable system.

[0046] The cooking system **200** can include: a communication module **210**, a power source **220**, optionally a user interface **230**, a cooking system connector **240**, a processing system **250**, and/or

any other suitable components.

[0047] The communication module **210** functions to enable communication between the cooking system **200** and a remote computing system **300** (e.g., transmit and/or receive data by the cooking system **200** to and/or from a remote computing system **300**). The cooking system **200** can include one communication module **210**, multiple communication modules **210** (e.g., same and/or different types), and/or any other suitable number of communication modules. The communication module **210** is preferably a wireless communication module (e.g., wireless module), but can additionally and/or alternatively be a wired communication module. The communication module **210** can be: Wi-Fi (e.g., WiFi module), Bluetooth (e.g., Bluetooth module; wherein the protocol can be Bluetooth classic, BLE, UWB, etc.), cellular, Zigbee, Z-wave, NFC, RF, mesh, radio, micro-USB, USB-A, USB-C, ethernet, and/or any other suitable communication module.

[0048] The communication module **210** preferably stores wireless credentials (e.g., network name, password, cryptographic key, WiFi credentials, Bluetooth credentials, etc.), but can alternatively not store wireless credentials (e.g., wherein the processing system **250** or other system can store the wireless credentials). The communication module **210** can connect to a local network and/or a remote computing system **300** using the wireless credentials, but can alternatively otherwise connect to a local network and/or a remote computing system **300**. For example, the communication module **210** is a communication system, such as that described in U.S. application Ser. No. 17/126,815 filed 18 Dec. 2020, which is incorporated in its entirety by this reference.

[0049] However, the communication module **210** can be otherwise configured.

[0050] The power source **220** functions to power the cooking system **200** and/or the temperature probe **100** for operation. The power source **220** is preferably a rechargeable battery, but can alternatively be a non-rechargeable battery (e.g., a disposal battery), induction coil, RF charger, wall-powered device, and/or any other suitable power source. The power source **220** can be: a lithium-ion battery, a nickel-cadmium battery, a nickel-metal-hydride battery, a lithium-ion polymer battery, and/or any other suitable rechargeable battery. The power source **220** is preferably charged via a connector (e.g., same or different as cooking system connector **240**), but can alternatively be charged via Bluetooth, induction, and/or any other suitable power charging component.

[0051] However, the power source **220** can be otherwise configured.

[0052] The user interface **230** functions to enable a user to interact with the cooking system **200**. The cooking system **200** can include: one user interface **230**, multiple user interfaces **230**, and/or any other suitable number of user interfaces. The user interface **230** preferably includes one or more user outputs, but can additionally and/or alternatively include one or more user inputs, and/or any other suitable component.

[0053] The user output is preferably a display, but can additionally and/or alternatively be: a projector, a speaker, a printer, a plotter, a headphone, an external system (e.g., wherein the output data is transmitted to the external system via the wireless connection), and/or any other suitable user output. The display functions to display information from the temperature probe **100**. The cooking system **200** can include: one display, multiple displays, and/or any other suitable number of displays. The information is preferably a temperature estimate determined by the processing system **250** of the cooking system **200**, but can additionally and/or alternatively be a food identifier, a target temperature, a timer, a temperature probe type, a temperature probe identifier, a temperature probe indicator (e.g., that a temperature probe **100** is connected to the cooking system **200**), a temperature probe-port association (e.g., which temperature probe and/or which type of temperature probe is connected to each port), a time elapsed, a time remaining, temperature estimates over a period of time (e.g., in a graph form, in a table form, etc.), temperature estimates aggregated over a period of time (e.g., minimum value, maximum value, average value, median value, mode value, etc.), and/or any other suitable information. The display can be a segmented LCD, a segmented LED, a smartwatch display, an OLED, and/or any other suitable display. Alternatively, the cooking system **200** can exclude a display.

[0054] The cooking system **200** can include: one user input, multiple user inputs, and/or any other suitable number of user inputs. The user input can be: a touchscreen (e.g., overlaid over the display, located in a different location, etc.), button(s), a keyboard, a mouse, gesture control, a microphone, a camera, and/or any other suitable user input. Additionally or alternatively, the cooking system **200** can lack user inputs (e.g., lack buttons, touchscreens, etc.), lack a mechanism for manual sensor probe type selection, and/or be otherwise limited.

[0055] However, the user interface **230** can be otherwise configured.

[0056] The cooking system connector **240** functions to connect the cooking system **200** to a temperature probe **100**. The cooking system connector **240** preferably removably connects the cooking system **200** to the temperature probe **100**, but can alternatively permanently connect the cooking system **200** to the temperature probe **100**, and/or be otherwise configured. The cooking system connector **240** can be: a mechanical connector, a signal connector, a data connector, a power connector, and/or any other suitable connector. The cooking system **200** can include: one cooking system connector **240**, multiple cooking system connectors **240** (e.g., 2, 3, etc.), and/or any other suitable number of connectors. The cooking system connector **240** is preferably electrically conductive, but can alternatively be electrically insulative, and/or have any other suitable properties. The cooking system connector **240** can be: metal (e.g., aluminum, stainless steel, copper, etc.), ceramic, polymer (e.g., silicone rubber, plastic, etc.), glass, and/or any other suitable material. The cooking system connector **240** is preferably a female connector (e.g., be a port), but can additionally and/or alternatively be a male connector. The cooking system connector **240** preferably connects to (e.g., receives) a male connector (e.g., probe connector **130** of the temperature probe), but can additionally and/or alternatively connect to a female connector. The cooking system connector **240** can be: a plug or socket connector, a crimp on connector, a soldered connector, a binding post, a screw terminal, a ring and spade connector, a blade connector, twist-on wire connector, alligator clip, and/or any other suitable connector. The cooking system connector **240** is preferably an auxiliary jack, but can additionally and/or alternatively be a USB-A port, a USB-C port, a micro-USB port, a headphone jack, and/or any other suitable connector. For example, cooking system connector **240** is an auxiliary jack that connects to probe connector **130**, which is an auxiliary plug.

[0057] The cooking system connector **240** preferably includes a set of contacts that electrically connect to the conductors of the probe connector **130**. The cooking system connector **240** preferably includes a contact for each conductor, but can alternatively include one contact for multiple conductors or multiple contacts for a single conductor. Each contact is preferably spatially indexed, such that the processor can determine which connector conductor is providing the signal; however, the contacts can be otherwise identified. The cooking system connector **240** can include: a number of contacts that correspond to a maximum number of possible conductors on probe connector **130** (e.g., 3, 5, 6, 7, etc.), a set number of contacts, and/or any other number of contacts. The cooking system connector **240** preferably conducts electrical signals provided by the probe connector **130** to the processing system, but can otherwise handle the electrical signals. However, the cooking system connector **240** can be otherwise configured.

[0058] The processing system **250** functions to control the cooking system **200** and/or temperature probe **100** operation. The cooking system **200** can include: one processing system **250**, multiple processing systems **250**, and/or any other suitable number of processing systems. The processing system **250** can include: one or more processors, a memory, and/or any other suitable component.

[0059] The processing system **250** can be configured to: identify the temperature probe based on the detected ground-signal pattern, determine the temperature resolution model for the identified temperature probe, convert temperature probe signals into temperature measurements and/or estimates, track temperature measurements over a time period, optionally operate the cooking system **200** and/or another system based on the temperature measurements, and/or perform any other suitable functionality. The temperature estimates can be stored onboard (e.g., in the memory

of the processing system **250**), stored remotely, and/or otherwise stored. The temperature estimates can be stored for: a predetermined timeframe, indefinitely, a food instance, a cook session, and/or any other suitable period of time.

[0060] The processing system **250** can be configured to store and/or execute one or more sensor resolution models from a set of models, and/or the temperature probe **100** is configured to be used with a set of models. Each sensor resolution model is preferably specific to a temperature probe type (e.g., 1-sensor RTD, 2-sensor RTD, 1-sensor NTC, 2-sensor NTC, 3-sensor NTC, etc.), but can additionally and/or alternatively be specific to a temperature probe class (e.g., NTC, RTD, etc.), a temperature sensor class (e.g., RTD, NTC), shared between all temperature probe types, a specific temperature probe instance, a cooking system **200**, and/or be otherwise specific or generic. Each model preferably converts an electrical signal (e.g., a resistance value, a voltage value, etc.) to a temperature estimate value, but can additionally and/or alternatively have any other suitable functionality. Each model can be: a machine learning model (ML model), a classical model, an equation, a lookup table, a graph, a database, a predetermined series of measurements, a predetermined series of analyses or transformations, a neural network, leverage regression, classification, rules, heuristics, instance-based methods (e.g., nearest neighbor), regularization methods (e.g., ridge regression), decision trees, Bayesian methods (e.g., Naïve Bayes, Markov, etc.), kernel methods, probability, deterministics, support vectors, and/or any other suitable model or methodology.

[0061] The models can be: generated onboard the cooking system, generated remotely (e.g., by the remote computing system **300**), programmed into the processing system **250**, updated via firmware updates (e.g., received from a remote computing system **300**), and/or otherwise determined. The models can be determined (e.g., learned, trained, etc.) based on: data from the remote computing system **300**, temperature estimates determined by the cooking accessory **200** and/or temperature probe **100**, user-assigned labels, and/or any other suitable training data. The models can be automatically determined, manually determined, predetermined, or otherwise determined. The models can be executed onboard the processing system **250**, remote from the cooking system **200** (e.g., by the remote computing system **300**), and/or by any other suitable system. The model can be executed using: electrical signals (e.g., resistance values, voltage values, etc.) from the temperature probe **100**, other measurements from the temperature probe **100**, measurements from the cooking system **200**, external data (e.g., from a database, from a user device, from a remote computing system **300**, etc.), and/or any other suitable data. For example, the model can convert a resistance measurement to a temperature value. In an illustrative example, a first model (e.g., for a first temperature probe) converts a resistance signal value to a first temperature value, and a second model (e.g., for a second temperature probe) converts the same resistance signal value to a second temperature value different from the first. However, models can be otherwise configured.

[0062] The processing system **250** can additionally or alternatively store and/or generate a set of fusion models, which function to summarize the measurements from multiple sensors. The measurements are preferably concurrent or contemporaneous, but can alternatively be from multiple timeframes. The fusion model to use can be: manually selected, automatically selected (e.g., based on the sensor type, temperature probe type, cooking system, cooking instructions, etc.), and/or otherwise determined. Examples of fusion models that can be used include: taking the mean of the measurements, median of the measurements, minimum of the measurements, mode of the measurements, maximum of the measurements, most stable measurement, the measurement with the least deviation, and/or any other model.

[0063] In a first example, a first fusion model is used to process multiple measurements from a first temperature probe type, and a different fusion model is used to process multiple measurements from a different temperature probe type. In a second example, a first fusion model is used to process multiple measurements from a first sensor type, and a different fusion model is used to process multiple measurements from a different sensor type (e.g., all NTC sensors are averaged

while all RTC measurements use floors). In a third example, the cooking instructions determine which fusion model to use, wherein the sensor resolution model is still selected based on the determined temperature probe type and/or sensor type. In a fourth example, a user selection determines which fusion model to use, wherein the sensor resolution model is still selected based on the determined temperature probe type and/or sensor type (e.g., whether to display a timeseries of spatial measurements according to the sensor's relative poses, or to fuse multiple readouts into a single value). However, the fusion models can be otherwise used.

[0064] The processing system **250** can be configured to: store and/or generate the one or more mappings between ground-signal conductor patterns and temperature probe types (e.g., identify a temperature probe based on the ground-signal conductor pattern); example shown in FIG. 7. The mappings are preferably programmed or loaded into the processing system **250** or memory, but can additionally and/or alternatively be generated onboard, generated remotely (e.g., by the remote computing system **300**), updated via firmware updates (e.g., received from a remote computing system **300**), and/or otherwise determined. The mappings are preferably stored in the processing system **250** (e.g., memory), but can additionally and/or alternatively be stored remotely (e.g., by the remote computing system **300**, by an external database, etc.).

[0065] The processing system **250** can be configured to: perform all or portions of the method (e.g., **S100-S300**). In variants, the processing system **250** can: receive a set of electrical signals from the temperature probe **100**, determine a ground-signal conductor pattern from the set of electrical signals, retrieve mappings between ground-signal conductor patterns and temperature probe types, determine a temperature probe type based on the determined ground-signal conductor pattern and the mappings, determine a model (e.g., a lookup table) associated with the temperature probe type, and determine a final temperature estimate using the model based on the set of electrical signals; example shown in FIG. 3.

[0066] The processing system **250** can be configured to: generate and/or present notifications. The notifications can be presented on: the user interface **230** (e.g., a display), an external display (e.g., a browser, a user device display, etc.), a smart speaker, and/or any other suitable interface. The notifications can be: visual, audio, haptic (e.g., vibration), and/or in any other suitable format. The notifications can be generated based on: a temperature estimate, an expected temperature estimate and/or range, a target temperature estimate, a food identifier, a timer, a set of predetermined ground-signal conductor patterns, and/or otherwise determined. In a first example, a notification is generated when the temperature estimate is not within an expected temperature estimate range. In a second example, a notification is generated when a temperature probe type is not identified (e.g., when a ground-signal conductor pattern determined based on the set of electrical signals does not match any of the set of predetermined ground-signal conductor patterns). In a third example, a notification (e.g., with the identified temperature probe type) is generated when the temperature probe type is identified.

[0067] The processing system **250** can be configured to: store calibration information and/or calibrate the cooking system **200** and/or temperature probe **100**. For example, the processing system **250** can store an electrical signal (e.g., measured signal) in association with a reference temperature. In a first specific example, a user is instructed to insert the temperature probe **100** into boiling water. The cooking system **200** (e.g., processing system **250**) can record the electrical signal measured by the temperature probe **100**, and store the electrical signal in association with 100° C. In a second specific example, a user is instructed to insert the temperature probe **100** into ice water. The cooking system **200** (e.g., processing system **250**) can record the electrical signal measured by the temperature probe **100**, and store the electrical signal in association with 0° C.

[0068] The processing system **250** can be configured to: control the cooking system **200** (e.g., based on a set of control instructions received from the remote computing system **300**, based on the temperature estimates, etc.). For example, the processing system **250** can execute a program associated with a set of control instructions (e.g., user instructions, automatic cooking instructions

for a food, etc.). The program can include: instructions to execute one or more models (e.g., a single model, a series of models, etc.), a predetermined analysis, a predetermined series of temperature estimates, a series of operation instructions (e.g., instructions to display a predetermined set of variables or visual assets, etc.), cooking element control instructions (e.g., to heat, steam, cool, and/or otherwise treat the food), and/or any other suitable program.

[0069] However, the processing system **250** can be otherwise configured.

[0070] However, the cooking system **200** can be otherwise configured.

[0071] The system can optionally include a remote computing system **300** (e.g., platform), which functions to generate models, store models, update the cooking system **200** with models, store temperature estimates, store user preferences, generate cooking instructions, and/or perform other functionalities. The remote computing system **300** can be accessible via an API, through a unique endpoint, through a unique uniform resource identifier (URI), and/or otherwise accessed. The cooking system **200** can connect to the remote computing system **300** via Wi-Fi, Bluetooth (e.g., using a user device such as a smartphone, tablet, laptop, and/or desktop), cellular, and/or any other suitable wireless communication, but can additionally and/or alternatively be accessible through a wired communication (e.g., Ethernet); example shown in FIG. 5. The remote computing system **300** can be: a cloud computing system, a distributed network, a server system, and/or any other suitable computing system. The remote computing system can include: a model training system, a model repository, data storage, a device database, and/or any other suitable component.

[0072] However, the remote computing system **300** can be otherwise configured.

4. METHOD

[0073] The method can include: determining a set of electrical signals **S100**, determining a temperature probe type **S200**, determining a final temperature estimate **S300**, and/or any other suitable elements.

[0074] All or portions of the method can be performed by a cooking system **200** (e.g., by the processing system **250**), a temperature probe **100**, a remote computing system **300**, and/or any other suitable system. All or portions of the method can be performed in real-time, near-real time, and/or otherwise performed. The method can be performed after a predetermined event (e.g., temperature probe **100** actuated), after receiving user input (e.g., cooking instructions), periodically, randomly, and/or any other suitable time.

4.1. Determining a Set of Electrical Signals **S100**.

[0075] Determining a set of electrical signals **S100** functions to determine one or more electrical signals from a temperature probe **100**. **S100** is preferably performed before **S200**, but can alternatively be performed concurrently with **S200**, after **S200**, and/or any other suitable time. **S100** is preferably performed by a processing system **250** of a cooking system **200**, but can additionally and/or alternatively be performed by a remote computing system **300**, and/or any other suitable computing system.

[0076] The set of electrical signals are preferably received from a temperature probe **100**, but can additionally and/or alternatively be received from an external system, and/or any other suitable system. The set of electrical signals can be received after a probe connector **130** of the temperature probe **100** connects to cooking system connector **240** of the cooking system **200**, after the temperature probe **100** is inserted into or thermally contacts an object, and/or at any other time. Each electrical signal of the set is preferably received from a conductor (e.g., a tip, a sleeve, a ring) of the probe connector **130** (e.g., via a contact of the cooking system conductor **240**), but can additionally and/or alternatively be received from a different component of the temperature probe **100**, a component of the cooking system **200**, and/or otherwise received. Electrical signals of the set are preferably concurrently and/or contemporaneously received, but can additionally and/or alternatively be serially received. The electrical signal is preferably associated with a resistance value, but can additionally and/or alternatively be associated with a voltage value, a current value, and/or any other suitable value.

[0077] However, the set of electrical signals can be otherwise determined.

4.2. Determining a Temperature Probe Type **S200**.

[0078] Determining a temperature probe type **S200** functions to identify a temperature probe type for the temperature probe **100** connected to the cooking system **200**. **S200** is preferably performed after **S100**, but can alternatively be performed concurrently with **S100**, before **S100**, and/or any other suitable time. **S200** is preferably performed by a processing system **250** of a cooking system **200**, but can additionally and/or alternatively be performed by a remote computing system **300**, and/or any other suitable computing system. In variants, **S200** described herein can be equally applicable to temperature sensor types.

[0079] The temperature probe type is preferably determined based a set of electrical signals (e.g., determined in **S100**), but can additionally and/or alternatively be determined based on a temperature probe identifier, retrieved from a database, selected manually (e.g., on user interface **230** of cooking system **200**), and/or otherwise determined.

[0080] In a first variant, the temperature probe type can be determined based on a ground-signal conductor pattern (e.g., spatial pattern) of the set of electrical signals. For example, unique ground-signal conductor patterns are preferably associated with different temperature probe types, but can additionally and/or alternatively not be associated with different temperature probe types. Each conductor (e.g., a tip, a ring, a sleeve, etc.) of the probe connector **130** can be associated with an electrical signal or is grounded.

[0081] In a first specific example, the temperature probe type is identified as a 1-sensor RTD when a tip of the probe connector **130** is grounded, an electrical signal is detected at a single ring of the probe connector **130**, and a sleeve of the probe connector **130** is grounded, wherein the probe connector **130** only includes a tip, a single ring, and a sleeve.

[0082] In a second specific example, the temperature probe type is identified as a 2-sensor RTD when a tip of the probe connector **130** is grounded, an electrical signal is detected at a first ring of the probe connector **130**, an electrical signal is detected at a second ring of the probe connector **130**, and a sleeve of the probe connector **130** is grounded, wherein the probe connector **130** only includes a tip, a first ring, a second ring, and a sleeve.

[0083] In a third specific example, the temperature probe type is identified as a 1-sensor NTC when an electrical signal is detected at a tip of the probe connector **130**, and a sleeve of the probe connector **130** is grounded, wherein the probe connector **130** only includes a tip and a sleeve.

[0084] In a fourth specific example, the temperature probe type is identified as a 2-sensor NTC when an electrical signal is detected at a tip of the probe connector **130**, an electrical signal is detected at a single ring of the probe connector **130**, and a sleeve of the probe connector **130** is grounded, wherein the probe connector **130** only includes a tip, a single ring, and a sleeve.

[0085] In a fifth specific example, the temperature probe type is identified as a 3-sensor NTC when an electrical signal is detected at a tip of the probe connector **130**, an electrical signal is detected at a first ring of the probe connector **130**, an electrical signal is detected at a second ring of the probe connector **130**, and a sleeve of the probe connector **130** is grounded, wherein the probe connector **130** only includes a tip, a first ring, a second ring, and a sleeve.

[0086] In a second variant, the temperature probe type can be determined based on a temperature probe identifier encoded within an electrical signal of the set of electrical signals. The temperature probe identifier can be: unique to the temperature probe instance (e.g., globally unique), unique to the temperature probe class (e.g., all temperature probes of the same make and/or model share the same identifier), nonunique, and/or otherwise related to other temperature probes. The temperature probe identifier can be used as the temperature probe type and/or be converted (e.g., by the processing system **250**) to the temperature probe type.

[0087] In a third variant, the temperature probe type can be selected by a user on a user interface **230** of a cooking system **200**, wherein the cooking system **200** is connected to the temperature probe **100**.

[0088] However, the temperature probe type can be otherwise determined. **S200** can optionally include providing the temperature probe type to a user interface **230** of the cooking system **200**. The user interface **230** is preferably a display as described above, but can additionally and/or alternatively be: a different display, a projector, a speaker, a printer, a plotter, a headphone, an external system, and/or any other suitable user output. For example, the temperature probe type is displayed on the display.

4.3. Determining a Final Temperature Estimate **S300**.

[0089] Determining a final temperature estimate **S300** functions to determine a final temperature estimate for an object using the temperature probe **100**. **S300** is preferably performed after **S200**, but can alternatively be performed concurrently with **S200**, before **S200**, and/or any other suitable time. **S300** is preferably performed by a processing system **250** of a cooking system **200**, but can additionally and/or alternatively be performed by a remote computing system **300**, and/or any other suitable computing system.

[0090] The final temperature estimate is preferably numerical, but can additionally and/or alternatively be categorical. The final temperature estimate is preferably continuous, but can additionally and/or alternatively be discrete. The final temperature estimate can be determined in Fahrenheit, Celsius, Kelvin, and/or any other suitable measurement unit. The final temperature estimate can be rounded to the nearest 0.001°, 0.01°, 0.1°, 1°, within any range bounded by any of the aforementioned values, and/or any other suitable rounding metric.

[0091] The final temperature estimate can be determined based on: a temperature probe type (e.g., determined in **S200**), a sensor type (e.g., determined in **S200**), a set of electrical signals received from the temperature probe (e.g., determined in **S100**; e.g., using a different instance of the electrical signal set; etc.), a cooking system **200**, and/or otherwise determined. The final temperature estimate is preferably determined using one or more sensor resolution models, but can additionally and/or alternatively be retrieved from a database, determined manually, and/or otherwise determined. The sensor resolution model is preferably determined (e.g., selected, retrieved, etc.) based on the temperature probe type (and/or sensor type) determined in **S200**, or be otherwise identified. The sensor resolution model is preferably retrieved from the processing system **250** (e.g., memory) of the cooking system, but can additionally and/or alternatively be retrieved from the remote computing system **300**, an external database, and/or otherwise determined. The model is preferably a lookup table (example shown in FIG. 8), but can additionally and/or alternatively be an equation, a regression model, a curve, any model as described above, and/or any other suitable model or methodology; example shown in FIG. 12.

[0092] In variants, determining a final temperature estimate can include using a model associated with the temperature probe type. In examples, **S300** can include determining a lookup table specific to the identified temperature probe type in **S200**, and retrieving a set of temperature estimates using the lookup table that corresponds to the set of electrical signals (e.g., resistance values, voltage values, etc.) determined in **S100**. The temperature probe **100** can include a single sensor **120** or multiple sensors **120**.

[0093] In a first specific example, the temperature probe **100** includes a single sensor **120**. **S300** includes retrieving a single temperature estimate that corresponds to a single electrical signal (e.g., resistance value) from the lookup table, wherein the single temperature estimate is used as the final temperature estimate.

[0094] In a second specific example, the temperature probe **100** includes multiple sensors **120**, wherein the multiple sensors **120** are the same, but located in different positions along the probe body **110**. In this example, **S300** can include: optionally determining a fusion model (e.g., based on the identified temperature probe type, sensor type, a cooking instruction, a user selection, etc.); retrieving multiple temperature estimates, each corresponding to multiple electrical signals (e.g., resistance values) from the lookup table; and determining a final temperature summary based on the multiple temperature estimates and optionally the fusion model.

[0095] However, the final temperature estimate can be otherwise determined.

[0096] **S300** can optionally include providing the final temperature estimate to a user interface **230** of the cooking system **200**. The user interface **230** is preferably a display as described above, but can additionally and/or alternatively be: a different display, a projector, a speaker, a printer, a plotter, a headphone, an external system, and/or any other suitable user output. For example, the final temperature estimate is displayed on the display.

5. EXAMPLES

[0097] In an example, the method can include: receiving a set of electrical signals from the temperature probe **100**, determining a ground-signal conductor pattern from the set of electrical signals, retrieving mappings between ground-signal conductor patterns and temperature probe types, determining a temperature probe type based on the determined ground-signal conductor pattern and the mappings, determining a sensor resolution model (e.g., a lookup table) associated with the temperature probe type, optionally determining a fusion model, and determining a final temperature estimate using the sensor resolution model and optionally the fusion model, based on the set of electrical signals; example shown in FIG. **9**. The temperature probe **100** can include: a probe body **110**, a set of sensors **120** (e.g., NTC, RTD, etc.), and a male probe connector **130**. The male probe connector **130** includes a set of conductors, wherein each conductor can be: a tip, a sleeve, or a ring.

[0098] Different processes and/or elements discussed above can be performed and controlled by the same or different entities. In the latter variants, different subsystems can communicate via: APIs (e.g., using API requests and responses, API keys, etc.), requests, and/or other communication channels.

[0099] Alternative embodiments implement the above methods and/or processing modules in non-transitory computer-readable media, storing computer-readable instructions that, when executed by a processing system, cause the processing system to perform the method(s) discussed herein. The instructions can be executed by computer-executable components integrated with the computer-readable medium and/or processing system. The computer-readable medium may include any suitable computer readable media such as RAMs, ROMs, flash memory, EEPROMs, optical devices (CD or DVD), hard drives, floppy drives, non-transitory computer readable media, or any suitable device. The computer-executable component can include a computing system and/or processing system (e.g., including one or more collocated or distributed, remote or local processors) connected to the non-transitory computer-readable medium, such as CPUs, GPUs, TPUS, microprocessors, or ASICs, but the instructions can alternatively or additionally be executed by any suitable dedicated hardware device.

[0100] Embodiments of the system and/or method can include every combination and permutation of the various system components and the various method processes, wherein one or more instances of the method and/or processes described herein can be performed asynchronously (e.g., sequentially), concurrently (e.g., in parallel), or in any other suitable order by and/or using one or more instances of the systems, elements, and/or entities described herein.

[0101] As a person skilled in the art will recognize from the previous detailed description and from the figures and claims, modifications and changes can be made to the preferred embodiments of the invention without departing from the scope of this invention defined in the following claims.

Claims

1. A method comprising: receiving, by a computing device from a temperature probe via a wireless connection, a first set of signals comprising a pattern specific to one or more sensors associated with the temperature probe; determining, based on the first set of signals, a type of the one or more sensors associated with the temperature probe; determining, based on the type of the one or more sensors associated with the temperature probe, a temperature model; receiving, from the

- temperature probe via the wireless connection, a second set of signals; determining, using the temperature model and based on the second set of signals, a temperature of a food as the food cooks; and causing the temperature of the food to be displayed.
2. The method of claim 1, wherein the computing device comprises a user interface for a cooking appliance.
 3. The method of claim 2, wherein the cooking appliance comprises a grill.
 4. The method of claim 1, wherein the pattern comprises a spatial ground-signal conductor pattern.
 5. The method of claim 1, wherein the one or more sensors associated with the temperature probe comprises at least one of: a negative temperature coefficient (NTC) thermistor; a resistance temperature detector (RTD); or a positive temperature coefficient (PTC) thermistor.
 6. The method of claim 1, wherein the computing device comprises a mobile computing device.
 7. The method of claim 1, wherein the determining the temperature of the food as the food cooks further comprises: determining one or more temperature estimates; and determine, based on the one or more temperature estimates, the temperature of the food.
 8. The method of claim 7, wherein the one or more temperature estimates is determined by: determining a set of resistance signal values based on the second set of signals; and converting the set of resistance signal values to the one or more temperature estimates using the temperature model.
 9. The method of claim 1, further comprising: receiving, by the computing device from a second temperature probe, a third set of signals; determining, based on the second set of signals, a second type of one or more sensors associated with the second temperature probe; determining, based on the second type of the one or more sensors associated with the second temperature probe, a second temperature model; receiving, from the second temperature probe, a fourth set of signals; determining, using the second temperature model and based on the fourth set of signals, a second temperature of a second food as the second food cooks; and causing the second temperature of the second food to be displayed concurrently with the temperature of the food.
 10. The method of claim 9, wherein the second type of the one or more sensors associated with the second temperature probe is different than the type of the one or more sensors associated with the temperature probe.
 11. A system comprising: a temperature probe configured to send a plurality of signals; and a computing device configured to: receive, from the temperature probe via a wireless connection, a first set of signals of the plurality of signals, wherein the first set of signals comprises a pattern specific to one or more sensors associated with the temperature probe; determine, based on the first set of signals, a type of the one or more sensors associated with the temperature probe; determine, based on the type of the one or more sensors associated with the temperature probe, a temperature model; receive, from the temperature probe via the wireless connection, a second set of signals of the plurality of signals; determine, using the temperature model and based on the second set of signals, a temperature of a food as the food cooks; and cause the temperature of the food to be displayed.
 12. The system of claim 11, wherein the computing device comprises a user interface for a cooking appliance.
 13. The system of claim 12, wherein the cooking appliance comprises a grill.
 14. The system of claim 11, wherein the pattern comprises a spatial ground-signal conductor pattern.
 15. The system of claim 11, wherein the one or more sensors associated with the temperature probe comprises at least one of: a negative temperature coefficient (NTC) thermistor; a resistance temperature detector (RTD); or a positive temperature coefficient (PTC) thermistor.
 16. The system of claim 11, wherein the computing device comprises a mobile computing device.
 17. The system of claim 11, wherein the computing device is configured to determine the temperature of the food as the food cooks by: determining one or more temperature estimates; and

determine, based on the one or more temperature estimates, the temperature of the food.

18. The system of claim 17, wherein the computing device is configured to determine the one or more temperature estimates by: determining a set of resistance signal values based on the second set of signals; and converting the set of resistance signal values to the one or more temperature estimates using the temperature model.

19. The system of claim 11, wherein the computing device is configured to: receive, from a second temperature probe, a third set of signals; determine, based on the second set of signals, a second type of one or more sensors associated with the second temperature probe; determine, based on the second type of the one or more sensors associated with the second temperature probe, a second temperature model; receive, from the second temperature probe, a fourth set of signals; determine, using the second temperature model and based on the fourth set of signals, a second temperature of a second food as the second food cooks; and cause the second temperature of the second food to be displayed concurrently with the temperature of the food.

20. The system of claim 19, wherein the second type of the one or more sensors associated with the second temperature probe is different than the type of the one or more sensors associated with the temperature probe.
